PR : 2 Robust Regression Engine

PART :- A

(Q1) What is Regularization in Machine learning? Why is it needed?

→ Regularization is a technique used in ML to reduce overfitting by adding a penalty term of the model's loss function. It controls the complexity of the model and prevents it from learning noise from training data.

When a model becomes too complex, it performs very well on training data but poorly on unseen data.

* Types of Regularization

L1 Regularization (Lasso)
L2 Regularization (Ridge)

(lambda) strength

→ Formula :

$$\text{Loss} = \text{Error} + \lambda \times \text{penalty}$$

Error → Prediction error

penalty → coefficient

Q2 Diffn B/W Ridge & Lasso Regularization ?

→

Ridge Regularization	Lasso Regularization
- Uses squared penalty	Uses absolute penalty.
- Reduces coefficient values	can reduce coefficients to zero.
- Keeps all features	performs feature selection.
- useful when all features matter.	useful when some features are unimportant

(Q3) What is cross validation and Why is it important.

→ Cross validation is a model evaluation technique where the dataset is split into multiple subsets. The model is trained on some folds and validated on the remaining fold repeatedly.

* Importance :-

- prevents overfitting
- gives reliable evaluation
- helps select best model
- Improves generalization

eg :- - Dataset divided into 5 parts
- model trains 5 times.

(Q4) Explain the following cross validation techniques :

(1) K- Fold Cross Validation :-

→ The dataset is divided into K equal parts.

<u>Steps</u> :- (1) Use K-1 folds for training
(2) Use remaining fold for validation
(3) Repeat K times
(4) Average performance

<u>Adv</u> :- Efficient, Reliable, commonly used.

(2) Stratified K Fold Cross validation :-

→ regression, traget values are binned before splitting.

purpose :-

- Maintain equal target distribution
- Better evaluation consistency.

<u>Adv</u> :- Balanced folds, Reduces bias.

Page No.
Date

(3) Leave - one - out cross validation (LOOCV) :-

→ Each observation is used as validation data once.

if dataset has N rows.
- Train N times
- 1 row used for testing each time.

Adv :- maximum training data usage

Dis adv :- very slow
computationally expensive

(4) Time series split :-

→ used for time dependent data.

Rule imp :- future data should never be used to predict past data.

Adv :- prevents data leakage
preserves chronological order.

Page No.
Date

Q5) Why are tree based models less sensitive to feature scaling?

→ Tree based models like :-

- Decision tree
- Random forest

split data using feature threshol instead of distance calculation

→ They do not depend on :-

- Euclidean distance
- Gradient magnitude

→ Therefore :-

- Scaling is not necessary.
- Raw values work effectively.